COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 5 : SLO 1 — Booth's Multiplication Algorithm (Setup)
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1. Fill in the Blanks (Remembering)

• The single bit register Q-1 is initially set to 0.
• The accumulator (A) and Q-1 are initialized to 0.
• The sequence counter (SC) is set to the number of BITS in the multiplier.
• The algorithm examines the last two bits of the Q and Q-1 registers to determine the next action.

2. True or False (Understanding)

• a. If Q0 Q-1 = '10', perform A = A - M → TRUE
• b. An arithmetic right shift is performed on A, Q and Q-1 in every cycle → TRUE
• c. The algorithm stops when the sequence counter becomes 1 → FALSE (it stops when SC becomes 0)
• d. If Q0 Q-1 = '01', perform A = A + M → TRUE
• e. Booth's algorithm is only effective for multiplying positive numbers → FALSE

3. Matching (Applying)

• a. 0 0 → 3. No operation
• b. 0 1 → 2. A = A + M
• c. 1 0 → 1. A = A - M
• d. 1 1 → 3. No operation

4. Step-by-Step Analysis (Analyzing): (+7) × (-3), M = 0111, Q = 1101, A = 0000, Q-1 = 0

• Step 1a: Q0 Q-1 = '10' → perform A = A - M: A = 0000 - 0111 = 1001.
• Step 1b: after arithmetic right shift → A = 1100, Q = 1110, Q-1 = 1.
• Step 2c: new Q0 Q-1 = '01' → perform A = A + M: A = 1100 + 0111 = 0011.
• Step 2d: after the second arithmetic right shift → A = 0001, Q = 1111, Q-1 = 0.

5. Algorithm Explanation (Creating)

• Booth's algorithm multiplies signed 2's-complement numbers efficiently by handling blocks of 1s in the multiplier. Instead of adding for every 1, it performs one subtraction at the start of a run of 1s (pattern 10) and one addition at the end (pattern 01).
• This reduces the number of arithmetic operations for long strings of 1s, and because it works directly on 2's-complement values it handles negative numbers with no extra sign logic, making multiplication faster and simpler in hardware.